

ORIGINAL ARTICLE

Interactive Small-Group Asthma Education in the Community Pharmacy Setting: A Pilot Study

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Background. This study aimed to compare the effects of two small-group asthma education interventions (one delivered by specially trained pharmacists (group A) and one delivered by a pharmacist researcher trained as an asthma educator (group B)) with usual care provided by community pharmacists (group C) on clinical and humanistic outcomes for people with asthma. **Methods.** Pharmacies were randomly selected to provide either group A, B, or C interventions. Data were collected at baseline, post intervention (groups A and B) and at 6 and 12 weeks (final visit). **Results.** Forty-eight people with asthma were recruited into groups A ($n = 16$), B ($n = 16$), and C ($n = 16$) and there were no significant differences between the groups at baseline. At 12 weeks there was a significant decrease in the proportion of patients with severe asthma/poor control in groups A and B compared with group C (56%, 44% and 50% to 25%, 13% and 50% [$n = 48$, $p < 0.05$], respectively). In Groups A and B, the proportion of patients with optimal metered dose inhaler (MDI) technique improved from 9% and 14%, respectively, at baseline to 82% and 93% ($n = 11$, $p = 0.02$, $n = 14$, $p < 0.001$), respectively, at 12 weeks. The proportion of patients with optimal dry powder inhaler (DPI) technique improved in Groups A and B from 0% and 8%, respectively, at baseline to 86% and 92% ($n = 7$, $p < 0.001$; $n = 13$, $p = 0.002$), respectively, at 12 weeks. No change in inhaler technique was observed for Group C. There were significant improvements in asthma knowledge scores in Groups A and B compared to Group C over time. **Conclusions.** Small-group asthma education delivered by pharmacists appears to be more effective than usual care in improving clinical and humanistic asthma outcomes.

Keywords patient education, small-group education, asthma education, community pharmacists, community pharmacy

INTRODUCTION

Asthma is a chronic, episodic respiratory condition that is primarily managed in the community (1). Patient education is one of the critical elements to successful asthma management emphasized in international and national asthma management guidelines (2, 3). Currently, asthma education and care are provided mostly in primary care by health care professionals, which include general practitioners, clinic nurses, and community pharmacists (4).

Although numerous randomized controlled trials utilising small group formats have been shown to be successful in secondary care settings (5), fewer studies using either individual (6–11) or small group formats (12–14) have been conducted in primary care settings and to date their success has not been established.

Community pharmacy provides a strategic venue for the provision of patient education about asthma. Traditionally, patient education provided by pharmacists has been individualized. However, group education has been shown to be as effective as individualized education with the added benefits of being simpler, more cost-effective and better received by patients and educators (15).

To date, no small-group asthma education provided by pharmacists in the community pharmacy setting has been studied. The aim of this study was to implement and compare

the effects of two small-group asthma education interventions, one pharmacist-led (group A) and one delivered by a pharmacist researcher trained as an asthma educator (group B) with usual care provided by community pharmacists (group C) on clinical and humanistic outcomes for people with asthma.

METHODS

Study Design

Ethics approval for the study was obtained from the Human Research Ethics Committee, University of Sydney and informed consent was obtained from all participants. The study was conducted between January and July 2005 and used a parallel group repeated measures design (Figure 1). Two geographically separated Sydney Metropolitan areas were selected to be the sites for this pilot study based on similarities in sociodemographic indicators and prevalence of asthma. Three pharmacies were randomly selected from these two distinct locations to provide either the education interventions group A and group B or the minimal intervention group C.

Recruitment and Training of Pharmacists

Pharmacists were recruited using a personal approach and pharmacists in groups A and B were trained in a half-day training workshop (facilitated by a health educator and experts in the area) supported by a comprehensive asthma manual. Asthma management and micro skills of teaching relevant to engaging active learning about asthma medication to small groups were the focus of the education. Group C pharmacists received no training. An interactive small-group asthma education program was delivered in the community

Ethics was approved in July 2004 (Ref 7558) by the Human Research Ethics Committee of the University of Sydney, Sydney, NSW 2006.

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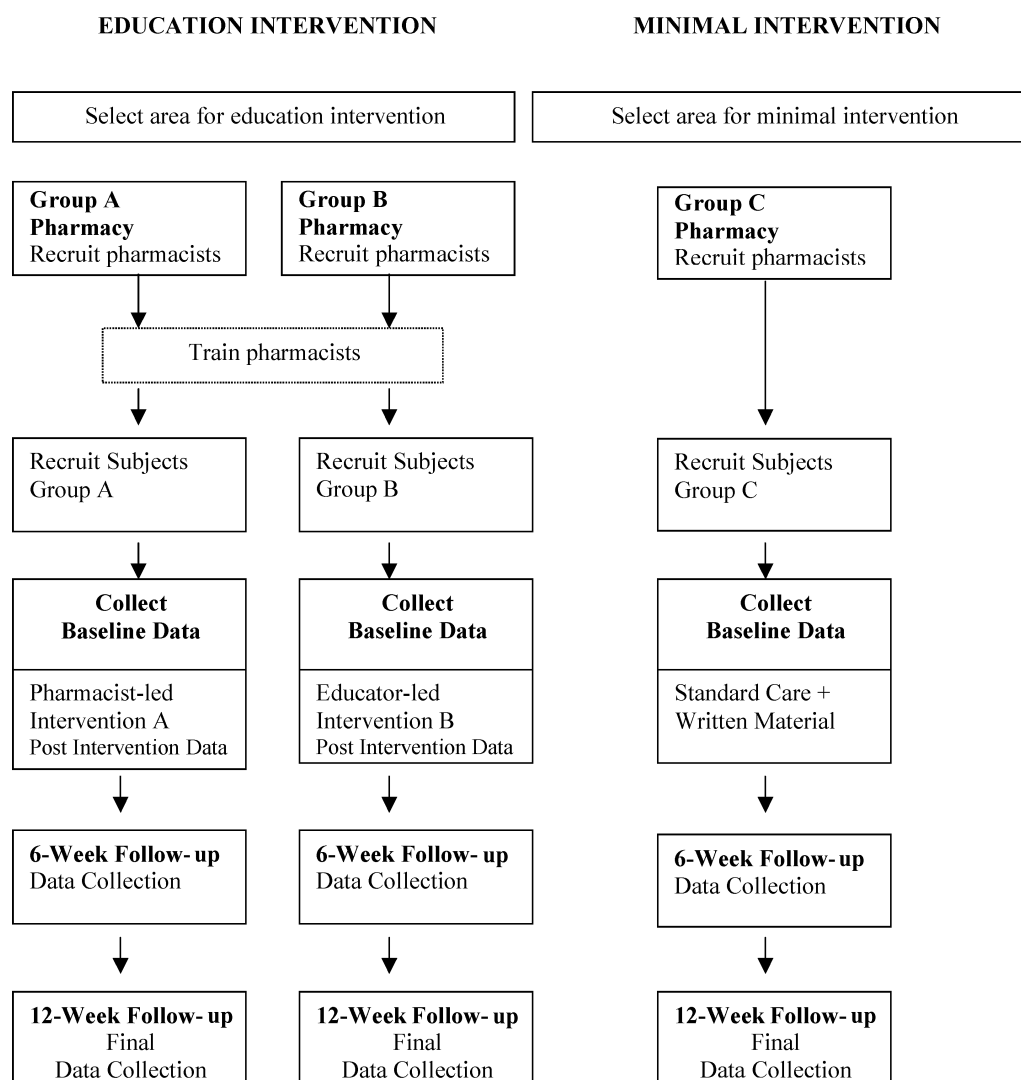


FIGURE 1.—Study design.

pharmacy after business hours by the trained pharmacists (group A) or the trained pharmacist who was also an asthma educator (group B). All pharmacists were provided with written information for participants and none were offered any remuneration.

Recruitment of Subjects

Inclusion criteria included people over 16 years of age with a previous self-reported clinical diagnosis of asthma who fulfilled at least one of the sub-criteria modified from the Revised Jones Morbidity Index for asthma assessment (16), who used preventer medication and who could read and understand English. People with self-reported clinical diagnosis of chronic obstructive pulmonary disease, those who were undergoing review with their general practitioner GP as part of the 3+ Visit Plan, or those who did not self-administer their inhaler were excluded.

Sample Size

To detect a change of 42% in the proportion of subjects who use their metered dose inhalers optimally (performed all

steps correctly) (17), with a power of 90% and significance level of 0.05, assuming a cluster effect of 1.5 and allowing for a subject dropout rate of 20%, 16 subjects were required to complete each of the three groups.

The Asthma Education Program

The intervention program delivered to small groups of participants (5–8) in groups A and B was a single interactive session of 150 mins duration focusing on the proper use of asthma medication. The session covered asthma, its management, asthma medication, inhaler use, and relevant written information (in the form of pamphlets and leaflets endorsed by the Asthma Foundation of New South Wales). Detailed program guidelines, (which included the use of an educational resource kit “Talk in A Box” provided by the Asthma Foundation of NSW), were prepared to guide pharmacists through each session and enable standardised delivery of the program. A checklist was used during the delivery of the education to audit the structure of the program. Group C participants did not receive specific asthma education but only received the same written information.

Data Collection

Data collected for all participants at baseline (first visit before any education) included demographics, asthma history, asthma severity/control, asthma medication, asthma action plan ownership, and concerns about asthma medication. The severity/control of asthma was categorized using a modified table from current Australian asthma management guidelines (3) and inhaler technique was assessed using a checklist for all possible devices. Optimal inhaler technique was recorded only if participants performed all the steps correctly for that device. The 5-item Medication Adherence Report Scale (MARS) (18) was used to assess participants' medication adherence, the 20-item Asthma Quality of Life Questionnaire (AQLQ) (19) was used to assess asthma-related quality of life, and asthma knowledge was assessed using the validated 12-item asthma knowledge questionnaire (CQ) (20). In addition, for participants in groups A and B, asthma knowledge and inhaler technique were assessed immediately after the education session.

Data collected for all participants at the 6-week follow-up included asthma severity/control, inhaler technique, medication adherence, asthma-related quality of life, asthma knowledge, and medication profiles. In addition, for participants in groups A and B, satisfaction regarding the education received was assessed using a modified participant satisfaction questionnaire from previous asthma studies (21). Similar data were collected at 12 weeks (final visit), including asthma action plan ownership and concerns about asthma medication.

Data Analysis

Data were analyzed using SPSS for Windows (version 10). Medications used were grouped under salbutamol, terbutaline, inhaled corticosteroids (ICS), combination therapy (ICS and a long-acting beta-agonist in one device), and anticholinergics. Optimal inhaler technique recorded for various devices were grouped into either optimal metered dose inhaler (MDI) technique or optimal dry powder inhaler (DPI) technique. Baseline categorical variables were summarized as percentages and associations tested in contingency tables by Chi-squared tests for independence. For continuous variables, which were normally distributed, analyses of variance (ANOVA) with Scheffe's post hoc comparisons were used to compare means between three independent groups. For continuous variables, which were not normally distributed, Kruskal-Wallis tests were used.

Student's paired *t* tests or Wilcoxon Signed Ranks tests were used to test for differences in asthma knowledge scores in each group pre- and post-delivery of asthma education. McNemar's tests were used to test for differences in the proportion of subjects who demonstrated optimal MDI and DPI technique in each group pre and post-delivery of asthma education. To test for differences in asthma knowledge scores between groups A and B immediately after the education, Student's *t* test or the Mann-Whitney U test was used. Multivariate repeated measures analysis was used to test for differences in mean scores within and between groups. Changes in categorical variables within each group over time were analyzed using Chi-squared tests for trend. A 2-tailed, 5% (0.05) level of significance was used for all statistical procedures.

RESULTS

Study Sample

The South East Sydney Area was selected as the site for the education interventions for group A and group B. The Central Sydney Area was selected as the site for the minimal intervention group C, as it matched most of the demographics.

Participation and Response Rates

All seven pharmacists approached participated in the study (3 in group A, 2 in each of groups B and C). Group A pharmacists delivered an asthma education program to three small groups ($n = 5$, $n = 5$, $n = 6$) of participants ($n = 16$). In group B, the asthma educator delivered the same program to three similar small groups ($n = 5$, $n = 6$, $n = 5$) of participants ($N = 16$). Educational sessions for participants in groups A and B were all delivered within 5 weeks after enrollment. Group C pharmacists enrolled 16 participants. Groups A, B, and C pharmacists performed follow-up visits with participants on an individual basis at 6 and 12 weeks. All 48 people with asthma from Groups A, B, and C completed the study (100% retention rate).

Baseline Characteristics

There were no statistically significant differences between the three groups at baseline in terms of demographic and asthma-related variables (Table 1).

There were no significant differences in medication use between Groups A, B, and C regarding salbutamol use (94%, 88%, and 75%), inhaled corticosteroid use (25%, 25%, and 19%), or combination therapy (75%, 75%, and 81%), respectively. Compared with other types of medication, the use of anticholinergics was low in all three groups. There were no significant differences in the proportion of subjects who owned a written asthma action plan and who had concerns about asthma medication (Table 1). In total, 24 subjects (50%) had concerns about their asthma medications, and steroid side effects were the main concerns reported.

On entry to the study there were no significant differences in mean MARS scores, total quality of life scores, median asthma knowledge scores, the proportion of subjects in each category of asthma severity/control, and the proportion of

TABLE 1.—Baseline characteristics of the three groups.

Profiles	Group A ($n = 16$) Mean $\pm$ SD	Group B ($n = 16$) Mean $\pm$ SD	Group C ($n = 16$) Mean $\pm$ SD	<i>p</i> value
Age (years)	49.5 $\pm$ 20.6	56.9 $\pm$ 19.4	46.4 $\pm$ 20.6	0.33
Male gender	5 (31)	6 (38)	9 (56)	0.33
Occupation				
Professional	5 (31)	2 (31)	5 (31)	
Non-professional	5 (31)	5 (31)	3 (19)	
Retired	3 (19)	5 (31)	5 (31)	0.90
Student	2 (13)	2 (13)	1 (6)	
Home duties	1 (6)	2 (13)	2 (13)	
Asthma-related conditions	6 (38)	11 (69)	10 (63)	0.20
Age at onset of asthma				
Infancy	2 (13)	2 (13)	4 (25)	
Childhood (2–12 years)	4 (25)	4 (25)	4 (25)	0.87
More than 12 years	10 (63)	10 (62)	8 (50)	
GP review in last 12 months	6 (38)	10 (63)	6 (38)	0.26
Action plan ownership	2 (13)	3 (19)	4 (25)	0.66
Concerns about medication	10 (63)	9 (56)	5 (31)	0.34

GP = general practitioner.

TABLE 2.—MARS, quality of life, asthma knowledge, inhaler technique, and asthma severity/control, categories at baseline.

Parameter (maximum score)	Group A (n = 16) Mean ± SD	Group B (n = 16) Mean ± SD	Group C (n = 16) Mean ± SD	Comparison between groups statistics
MARS (25)*	18.3 ± 4.0	19.5 ± 4.6	19.9 ± 3.2	F = 0.78; df = 2; p = 0.46
Total quality of life (10)†	2.5 ± 1.4	2.8 ± 1.6	2.3 ± 1.8	F = 0.27; df = 2; p = 0.76
Asthma knowledge (12)**	6.0(4.5–7.5)	6.5(5.5–7.5)	7.5(6.6–8.4)	$\chi^2 = 4.77$; df = 2; p = 0.09
Asthma severity/control category				
Mild	1(6)	1(6)	1(6)	$\chi^2 = 0.54$; df = 4; p = 0.97
Moderate	6(38)	8(50)	7(44)	
Severe	9(56)	7(44)	8(50)	
Optimal MDI technique	1, n = 11 (9)	2, n = 14 (14)	1, n = 11 (9)	$\chi^2 = 0.23$; df = 2; p = 0.89
Optimal DPI technique	0, n = 7 (0)	1, n = 13 (8)	2, n = 13 (15)	$\chi^2 = 1.35$; df = 2; p = 0.51

*A higher score indicates higher levels of reported adherence to medication.

†A higher score indicates a greater impact of asthma on quality of life.

**Median (IQR) score.

MARS = Medication Adherence Report Scale; MDI = metered dose inhaler; DPI = dry powder inhaler.

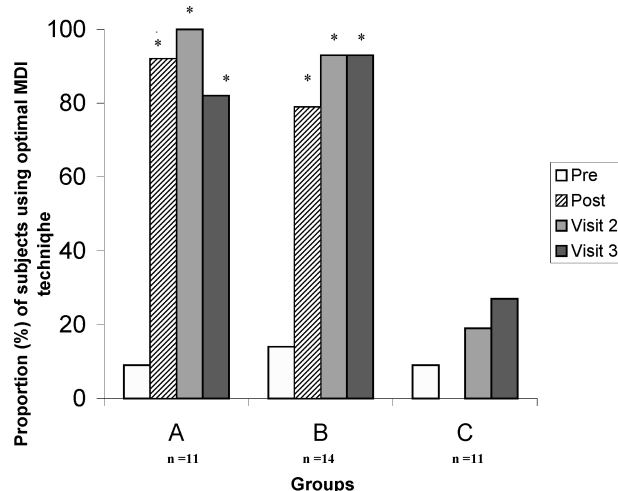
subjects who demonstrated optimal MDI and DPI technique between the three groups (Table 2).

Impact of Education

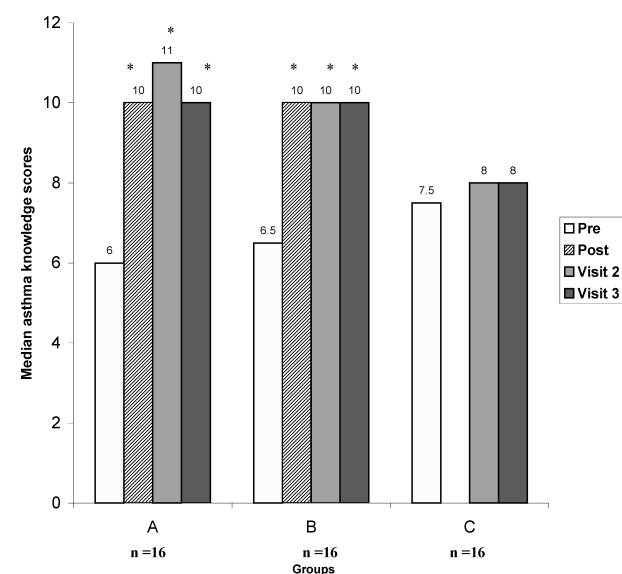
Asthma Knowledge. Figure 2 shows the median asthma knowledge scores at baseline, post-education, at 6 and 12 weeks. There was a statistically significant but equal increase in the median asthma knowledge scores in group A ($p < 0.001$) and in group B ($p < 0.001$) immediately after the education. This increase was sustained over the 12 weeks of the study for both groups. Knowledge scores were statistically significantly higher for groups A and B when compared to group C at 6 weeks ($p < 0.001$, $p < 0.001$, respectively) and at 12 weeks ($p < 0.001$, $p < 0.001$, respectively).

Inhaler Technique

Optimal metered dose inhaler (MDI) technique. Figure 3A shows the proportion of participants who demonstrated

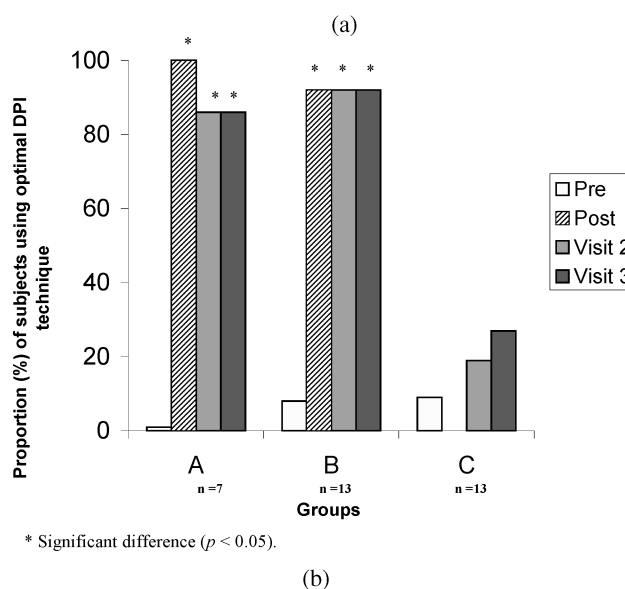


* Significant difference ($p < 0.05$).



* Significant difference ($p < 0.001$).

FIGURE 2.—Median asthma knowledge scores for group A (pharmacist-led), group B (asthma educator-led), and group C (usual care) at pre and post education, at 6 weeks (visit 2) and 12 weeks (visit 3). *Significant difference ($p < 0.001$).



* Significant difference ($p < 0.05$).

FIGURE 3.—The proportion (%) of subjects demonstrating optimal MDI technique in each group at pre and post education, at 6 weeks (visit 2) and 12 weeks (visit 3). *Significant difference ($p < 0.05$). (b) The proportion (%) of subjects demonstrating optimal DPI technique in each group at pre and post education, at 6 weeks (visit 2) and 12 weeks (visit 3). *Significant difference ($p < 0.05$).

optimal MDI technique at baseline, post-education, at 6 and 12 weeks. There was a significant improvement in group A ($p = 0.004$) and in group B ($p = 0.004$) immediately after the education, which was not statistically different ($p = 0.78$). This improvement was sustained over the 12 weeks of the study for both groups. In groups A and B, the proportion of patients with optimal MDI technique improved from 9% and 14%, respectively, at baseline to 82% and 93% ($n = 11$, $p = 0.02$; $n = 14$, $p < 0.001$), respectively, at 12 weeks.

Optimal dry powder inhaler (DPI) technique. Figure 3B shows the proportion of participants who demonstrated optimal DPI technique at baseline, post-education, at 6 and 12 weeks. There was a significant improvement both in group A ($p < 0.001$) and in group B ($p = 0.002$) immediately after the education that was not statistically different ($p > 0.5$). This improvement was sustained over the 12 weeks of the study for both groups. The proportion of patients with optimal DPI technique improved in groups A and B from 0% and 8%, respectively, at baseline to 86% and 92% ($n = 7$, $p < 0.001$; $n = 13$, $p = 0.002$), respectively, at 12 weeks.

Asthma Severity/Control. Figures 4A, 4B, and 4C show the proportion of subjects in each asthma severity/control category at baseline at 6 and 12 weeks for groups A, B, and C respectively. There were significant changes in the proportion of subjects in each severity/control category over time in group A ($p = 0.003$) and group B ($p = 0.001$) but not in group C. The difference in change from baseline between the three groups was significant at 6 weeks ($p = 0.05$) and at 12 weeks ($p = 0.02$). At 12 weeks, the proportion of subjects in the severe asthma/poor control category in group A (25%) and B (13%) were not significantly different ($p = 0.65$) but were significantly lower than group C (50%) ($p = 0.04$, $p = 0.006$, respectively).

Asthma-Related Quality of Life. Table 3 summarizes the total asthma quality of life (AQOL) scores at baseline at 6 and 12 weeks for the three groups. The impact of asthma on quality of life was modest for each group on entry into the study. There was a significant decrease in mean total AQOL scores over time in group A ($p = 0.03$) and group B ($p = 0.003$), indicating an improvement in quality of life, which was significantly different at 12 weeks ($p = 0.02$). At 12 weeks, group B had a significantly lower mean AQOL score compared to group C ($p = 0.02$).

Medication Adherence. Table 4 summarizes the MARS scores at baseline, at 6 and 12 weeks for the three groups. Relatively high MARS scores were reported at baseline. Although there were significant improvements in mean MARS scores over time in group A ($p < 0.001$) and group B ($p = 0.03$), there were no differences between the three groups at 12 weeks ($p = 0.52$).

Other Asthma-Related Variables. There was a significant decrease in the proportion of subjects who had concerns about their asthma medications from baseline to final visit in group A (63% to 25%) ($p = 0.03$) and group B (56% to 13%) ($p = 0.01$) but not in group C (31% to 25%) ($p = 0.69$).

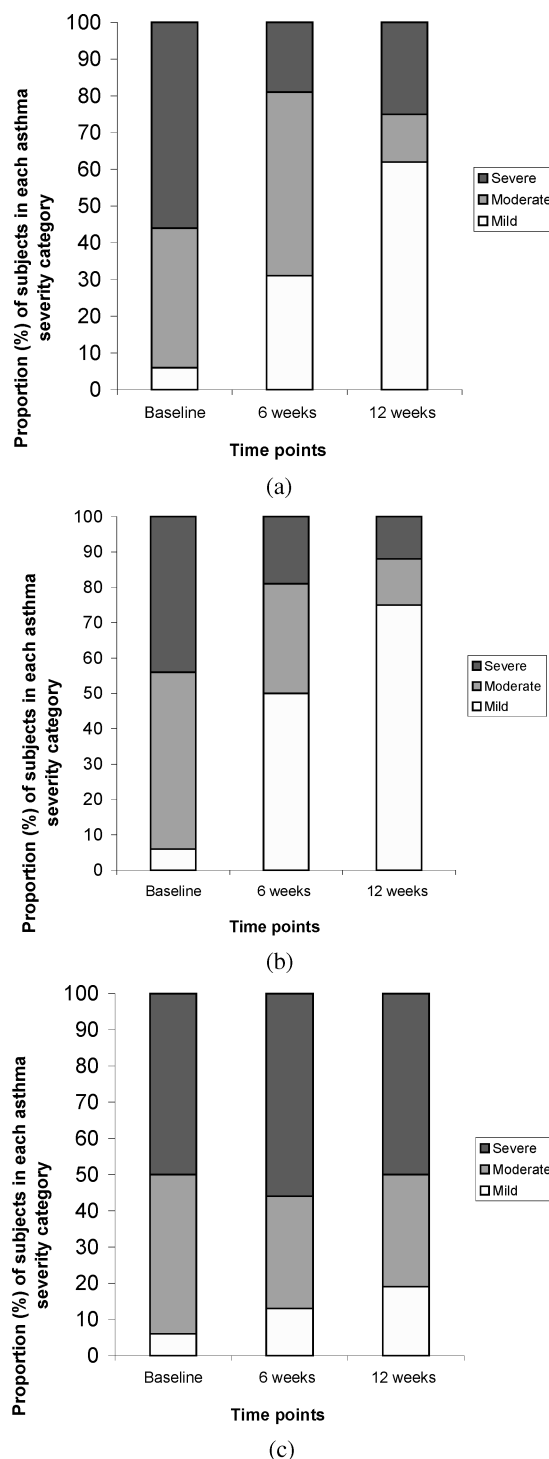


FIGURE 4.—(a) The proportion (%) of subjects in each asthma severity/control category at baseline, 6 weeks (visit 2), and 12 weeks (visit 3) for group A ($n = 16$). (b) The proportion (%) of subjects in each asthma severity/control category at baseline, 6 weeks (visit 2), and 12 weeks (visit 3) for group B ($n = 16$). (c) The proportion (%) of subjects in each asthma severity/control category at baseline, 6 weeks (visit 2), and 12 weeks (visit 3) for group C ($n = 16$).

The difference in change from baseline to final visit between the three groups was not significant ($p = 0.6$). There were no significant differences in medication use or action plan ownership between the three groups at the final visit.

TABLE 3.—Total asthma quality-of-life (AQOL) scores for the three groups.

Total asthma quality of life (maximum score = 10)	Group A (n = 16) Mean (95% CI)	Group B (n = 16) Mean (95% CI)	Group C (n = 16) Mean (95% CI)	Comparison between groups p value
Baseline*	2.5 (1.8 to 3.2)	2.8 (1.9 to 3.6)	2.3 (1.4 to 3.3)	0.76
6 weeks (visit 2)*	1.5 (1.1 to 1.9)	1.3 (0.8 to 1.9)	2.2 (1.3 to 3.1)	0.14
12 weeks (visit 3)*	1.4 (1.0 to 1.8)	1.0 (0.4 to 1.6)	2.3 (1.4 to 3.2)	0.02

*A higher score indicates a greater impact of asthma on quality of life.

Participant Satisfaction. Based on the participant satisfaction questionnaire completed at 6 weeks all 32 participants (groups A and B) indicated that they had been satisfied with the asthma class they had attended (in terms of content and time, pharmacists' knowledge, skills and ability to answer questions), they felt comfortable being educated in a group session and that they had benefited from the social support provided by the group. Thirty-one subjects (97%) indicated that the scheduled time and pharmacy setting for the class were suitable, indicating they would participate in a similar asthma class in the future and would recommend the class to a friend with asthma. The majority found many aspects of the asthma education program very useful, in particular how their medications worked, the rationale for their use, and how to use their asthma devices correctly. Twenty-five subjects (78%) indicated that they had not received any previous asthma education and those who had received it found this education better than that previously received.

DISCUSSION

The results of this study demonstrate that community pharmacist-led asthma education that focuses on the proper use of asthma medication, delivered to small groups of people with asthma, appears to be more effective than usual care in improving asthma severity/control, inhaler technique, and asthma knowledge. Specially trained community pharmacists can be as effective as a pharmacist researcher trained as an asthma educator in delivering small-group asthma education. This is the first time in Australia that the outcomes associated with such an innovative role in pharmacist-led small-group asthma education have been evaluated and indicated that the pharmacist educators have been pioneers in demonstrating the feasibility and efficiency of this role in the community pharmacy setting.

Our study is consistent with previous studies in demonstrating the value of a small group asthma education programs for adults (12, 13, 22–26). Although evidence suggests that both small group and individual delivery formats can improve patient health outcomes (e.g., reduction in emergency department and hospital visits), small-group education can be more effective for certain outcomes (e.g., improvement in inhaler technique) (25) with the added benefit of being simpler and

more cost-effective than individualized education (15). In our study, the efficiency of this method of patient education was demonstrated by the minimal training required.

The asthma education developed in this project was based on a theoretical framework (27), a model for health promotion planning and evaluation (28), by drawing on the literature (5) and incorporating the latest practice guidelines used in Australia (3). Evidence shows that asthma education programs are more likely to be successful if they are based on a theory of patient behavior change and focus on improvements beyond patient knowledge (29). The small-group asthma education program was carefully designed to focus on the proper use of asthma medication given that the major problems in asthma management are patient issues associated with medication use, an area in which pharmacists have expertise. It has been well documented that people with asthma are reluctant to use their asthma medication as recommended (30), do not optimally use their inhalers (31), and have unmet information needs about their asthma medication (30). Pharmacists were able to address all these issues during the asthma education program, deliver messages that were consistent with best practice guidelines and practitioners' instructions, and were in a pivotal position to reinforce key messages at the patient's next regular pharmacy visit.

In response to the small group asthma self-management education, there were improvements in participants' inhaler technique and asthma knowledge as shown previously in other similar group education studies (13, 17, 23–26). For improvements in inhaler technique to be achieved, previous studies have shown that written instructions alone are not sufficient and that verbal instructions, demonstrations, and practice sessions need to be included (25–32). Although short-term improvements in knowledge and inhaler technique are rarely sustained without ongoing review and reinforcement (33), pharmacists are ideally placed to review and reinforce correct inhaler technique during follow-up visits. In fact, we know that pharmacists can sustain improvements in inhaler technique (32).

This study showed an improvement in asthma severity/control in the education intervention groups. It is likely that the improvement in inhaler technique contributed to the improvement in asthma severity/control as there

TABLE 4.—MARS scores for the three groups.

MARS (maximum score = 25)	Group A (n = 16) Mean (95% CI)	Group B (n = 16) Mean (95% CI)	Group C (n = 16) Mean (95% CI)	Comparison between groups p value
Baseline*	18.3 (16.1 to 20.4)	19.5 (17.1 to 21.9)	19.9 (18.3 to 21.6)	p = 0.46
6 weeks (visit 2)*	21.3 (19.1 to 23.5)	22.1 (20.3 to 23.9)	20.1 (17.9 to 22.3)	p = 0.36
12 weeks (visit 3)*	21.1 (19.5 to 22.7)	21.9 (19.7 to 24.0)	20.6 (19.2 to 21.9)	p = 0.52

*A higher score indicates higher levels of reported adherence to medication.

were no significant differences in medication adherence and medication profiles reported at the final visit. Fewer concerns about asthma medication reported by the education groups at the final visit could also be a contributor to improvements in asthma severity/control as it has been shown that people with asthma have fears and many misconceptions regarding inhaled corticosteroids and their side effects, reducing their willingness to use them (30). Thus, both improved inhaler technique and willingness to take medications may have contributed to the improved asthma severity/control that we observed.

In the primary care setting, previous asthma self-management education studies have had a positive impact on patient's asthma quality of life (10, 11), however, these studies used individual delivery formats. In our study, a significant improvement in asthma quality of life was observed only in the asthma educator-led group. A possible explanation for this finding was that by the end of the study, 25% of subjects in the pharmacist-led group were categorized with severe asthma/poor control compared to the 13% in the asthma educator-led group, and although these proportions were not significantly different, the impact of asthma on quality of life was significantly smaller in the asthma educator-led group. Thus, asthma quality of life was a more sensitive measure than asthma severity/control. Furthermore, at baseline, asthma quality-of-life scores were low and self-reported medication adherence (MARS) scores were high, leaving small room for improvements to occur in these outcome measures within the study time frame.

The participant attendance and retention rates of 100% seem inconsistent with patterns observed in similar studies using a group format in various health care settings (22–24, 34). Generally, people with asthma are known to have poor attendance rates and low retention rates at group asthma education events. The high attendance and retention rates in our study indicated a committed and motivated sample, recruited to participate in a familiar local setting, involving one intensive education session with two less intensive visits within a short follow-up period. The fact that subjects were not randomly selected could have had an impact on the attendance and retention rates. Whether the pharmacist/patient relationship or the nature of the intervention was the main factor in the high retention rates was not explored.

The asthma education provided by pharmacists was perceived by participants as very useful, and both participants and pharmacists alike were highly satisfied. It has been suggested that if the health care provider is credible, the recommendations are practical, and the patient is satisfied with the patient/provider relationship, the patient may better adhere with the treatment and may have improved asthma outcomes (35). Participants considered the most useful advice they received was on how their medications worked and the rationale for their use and how to use their asthma devices correctly. Such features associated with appropriate medication use are highlighted in the current asthma management guidelines and lie at the core of existing problems faced in asthma management in Australia. Pharmacists may be inclined to assume that people with asthma have already been well educated at initial diagnosis or thereafter by other members of the health care team. In this study, 25 participants (78%) indicated that they had not received any previous

asthma education. Thus, a gap exists here, which provides an opportunity for pharmacists to fill, in their contribution to the overall management of asthma.

The results from this study may be relevant to other chronic disease states and further add to the body of evidence that pharmacists can have an impact on clinical and humanistic outcomes for people with asthma and potentially reduce health care utilization as shown previously through other more intensive community pharmacy-based asthma services in Australia and in Europe (21, 36).

There were several limitations to this pilot study, including sampling bias, selection bias, small sample size, and short follow-up, all of which affect the ability to generalize the results. However, this study forms the framework on which future studies should be designed and tested. Thus, the long-term effectiveness and sustainability of this model of pharmacist-led asthma education may be evaluated.

CONCLUSION

Small-group asthma education delivered by specially trained pharmacists appears to be more effective than usual care in improving clinical and humanistic outcomes for people with asthma. Specially trained community pharmacists are as effective as a pharmacist researcher trained as an asthma educator in delivering small-group asthma education.

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